Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/27/2006
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Project Title:	Development and Application of Reliability Analysis Techniques for Early Advanced Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/30/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HG90G		
Performance Goal No.:			
Performance Goal Text:			
	The integration of bioregenerative and physicochemical life support components has been achieved only in a limited basis on small-scale applications. When such systems are scaled to larger proportions and are pushed to extremes in the workable envelope the system reliability needs to be understood. Providing answers to the following fundamental questions concerning the reliability of ALS systems will be the research agenda of this proposed project: · What are the failure modes of an ALS system? · How reliable is a particular ALS system design? · How sensitive is the reliability of a system to the variability in the system? · How much flexibility in process level design will be desirable in an ALS system? · How sensitive is the reliability of a system to changes in its configuration? · To what extent will the integrated hybrid physicochemical and bioregenerative configurations become favorable? · What form of redundancy achieves acceptable reliability? · What amount of preventive and corrective maintenance is necessary? · What is the effect of designing reliable systems on ESM? · How can changes in system design have positive effects on		

Task Description:	both reliability and ESM? This research work proposes to superimpose estimated and assumed probabilistic distributions onto system components in an existing overall system model. This will afford the ability to simulate the stochastic nature of the system and ultimately determine the rate and manner in which the system may fail. A Monte Carlo analysis will be performed with the new stochastic overall system model, providing a comprehensive analysis of the state of system components when the system is both operational and failed. This information can be utilized to analyze the failure modes of a system, determine reliability, sensitivity, experiment with system configurations, schedule maintenance, and determine the correlation of reliability with ESM. At this stage, probabilistic information describing system components is limited and no comprehensive physical model of ALS system exists. However, this is an opportune time to use predictive systems analysis methodologies to study the important issues of ALS system reliability. Therefore, an integrated systematic approach, employing the techniques of ALS system modeling, probabilistic and stochastic methods, and reliability theories will be taken in this proposed project. This approach will also provide a mechanism for readily incorporating realistic process level reliability data for systems level reliability analysis when such data becomes available. For example, in a collaborative research effort, the generation of process level reliability information at the early stage of research is being planned. The analysis of component failure rates will be applied to a wastewater processing system as a test case. The use of the results from this collaborative research will be an ideal way to demonstrate the utility of the system reliability analysis tools and techniques developed in this proposed project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated:)